12 In an experiment 500 fruit flies were released into a controlled environment. After 10 days there were 650 fruit flies present.

Munirah believes that N, the number of fruit flies present at time t days after the fruit flies are released, will increase at the rate of 4.4% per day. She proposes that the situation is modelled by the formula $N = Ak^t$.

(i) Write down the values of A and k. **[2]**

(ii) Determine whether the model is consistent with the value of N at $t = 10$. **[2]**

(iii) What does the model suggest about the number of fruit flies in the long run? **[1]**

Subsequently it is found that for large values of t the number of fruit flies in the controlled environment oscillates about 750. It is also found that as t increases the oscillations decrease in magnitude.

Munirah proposes a second model in the light of this new information.

$$N = 750 - 250 \times e^{-0.092t}.$$

(iv) Identify three ways in which this second model is consistent with the known data. **[3]**

(v) (*A*) Identify one feature which is not accounted for by the second model. **[1]**

(*B*) Give an example of a mathematical function which needs to be incorporated in the model to account for this feature. **[1]**

12(i)	
12(ii)	

12(iii)	
12(iv)	
12(v)(*A*)	
12(v)(*B*)	

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